

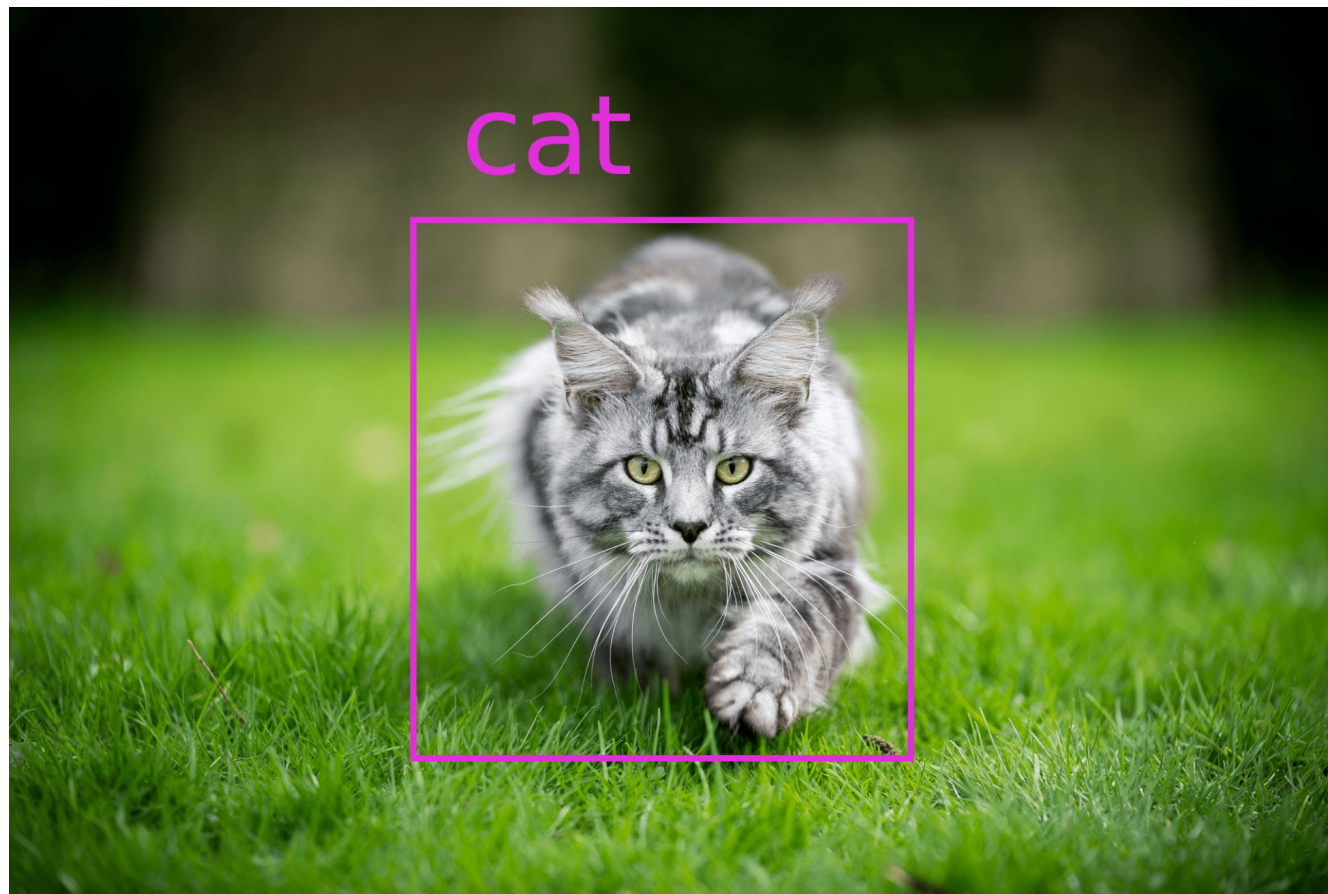


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OBJECT LOCALISATION AND OBJECT DETECTION



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Object Localization and Object Detection

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 - Sliding Window Approach
 - Regional Based Networks
 - YOLO

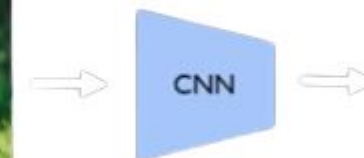
Object Localization and Object Detection

What is Object Localisation?

- **Object localisation is finding what and where a single object exists in an image.**

This involves 2 steps

- **Object Classification-** to identify the object in the image
- **Finding the location-** to find coordinates of the bounding box

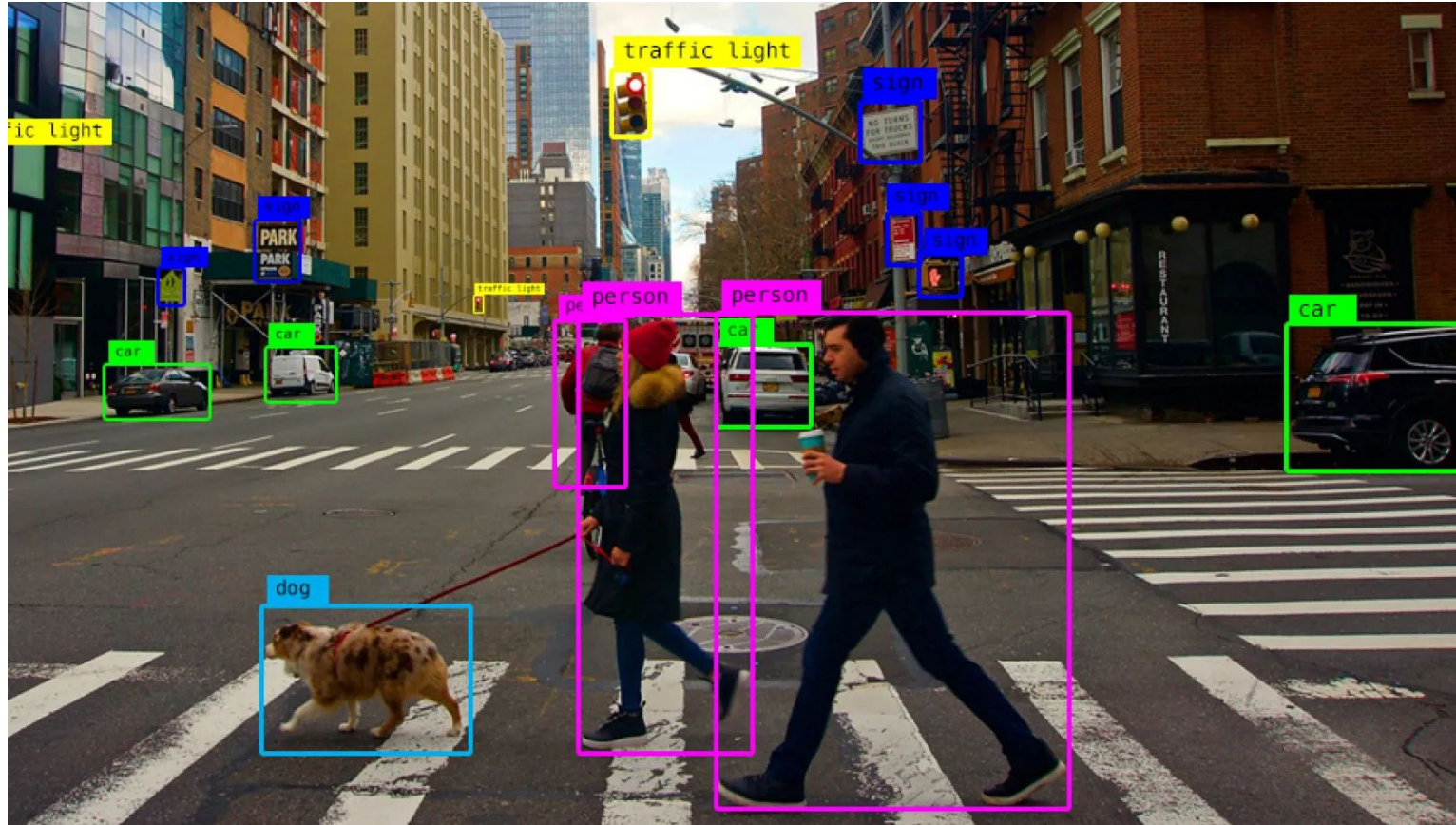


- **All this information is stored in a vector.**

Object Localization and Object Detection

What is Object Detection?

- **Object detection is detecting and localising multiple objects in an image.**
- **This involves classifying each object and predicting its bounding box.**





Object Localization and Object Detection

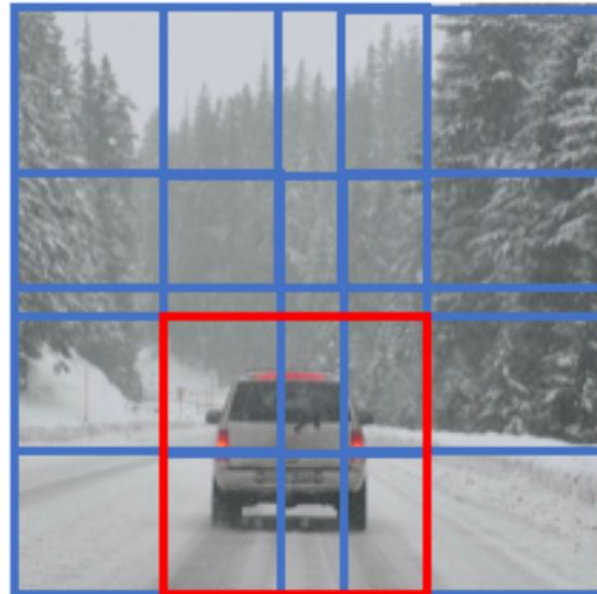
3 Approaches to Object Detection

- **Sliding Window Approach**
- **Regional based networks**
- **YOLO**

Object Localization and Object Detection

Sliding Window Approach

- **Moving a fixed size window across the image at different scales and positions to detect objects by classifying each window as containing an object or not .**
- **Disadvantage:**
It is computationally expensive and slow- as it processes numerous windows, many of which are redundant or irrelevant.



Object Localization and Object Detection

Regional Based Networks (R-CNNs)

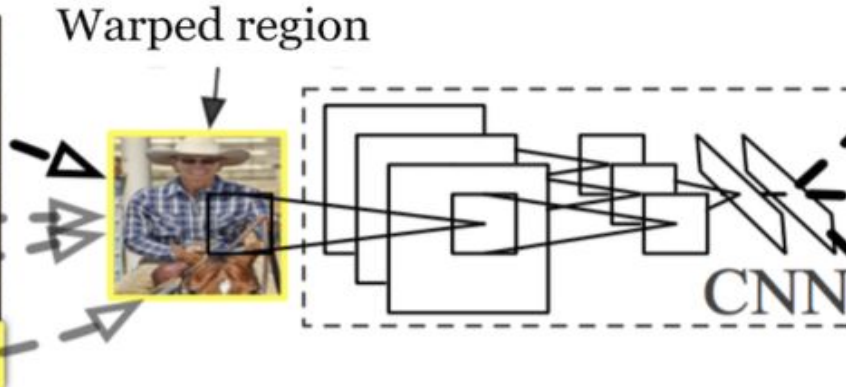
- **These models first generate region proposals (potential object locations) and then classify each proposal using a CNN.**
- **Disadvantages:**
 - **Slow Training and Inference-** Since it is a two-step process it is slower than single-shot detectors like YOLO
 - **Memory Intensive-** Requires extra memory to store region proposals



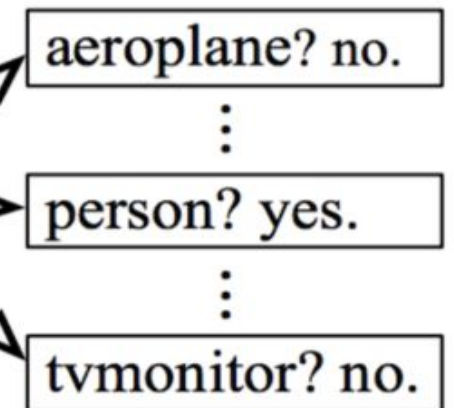
1. Input images



2. Extract region proposals (~2k)



3. Compute CNN features



4. Classify regions



Object Localization and Object Detection

YOLO

YOLO (You Only Look Once) is a fast and efficient approach for object detection, designed to improve on models like R-CNN.

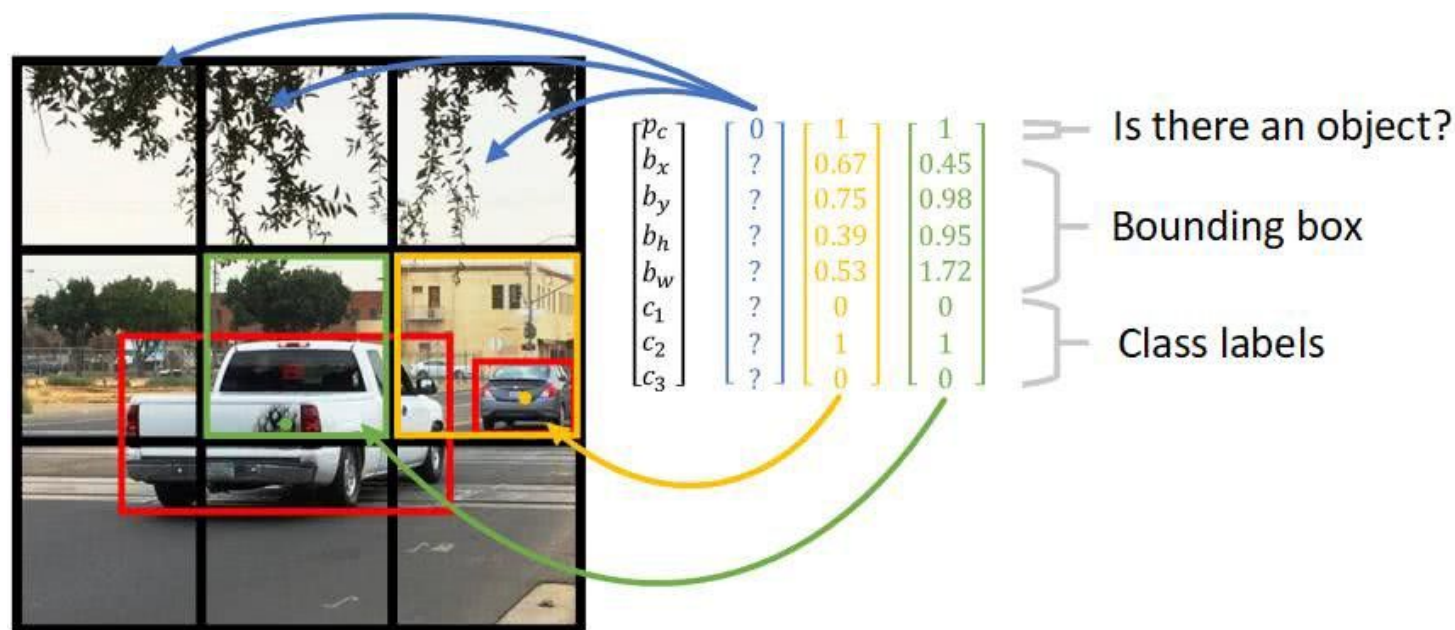
How YOLO Works-

- **YOLO divides the entire image into a grid (instead of generating separate region proposals)**
- **Each grid cell is responsible for detecting objects that fall within it.**
- **For each grid cell, the model predicts:**
 - **Bounding boxes (location of objects).**
 - **Class labels (what object it is, like cat, dog, etc.).**
 - **Confidence score (how sure it is about its prediction).**

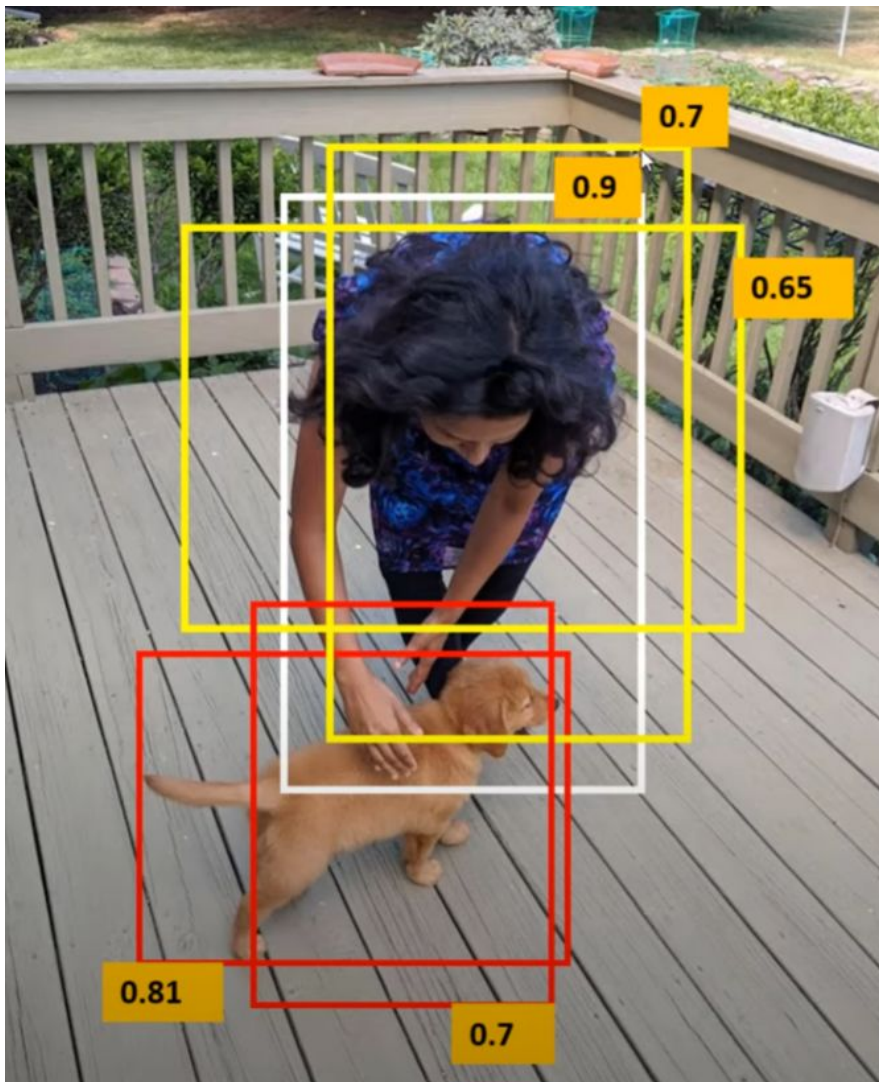
Object Localization and Object Detection

Advantages-

- **Single pass- hence extremely fast**
- **Fewer computations- Does not need region proposals and CNN for each box like R-CNN, hence is computationally efficient.**
- **Real-time Detection is possible- Since it processes the image in one go, it can be used for applications like video surveillance and self-driving cars.**



Object Localization and Object Detection



- **Problem-** there could be multiple bounding rectangles for 1 object.
- **We cannot take the max probability-** because there could be 2 objects very close to each other which contribute to the probability.
- **Intersection over union (IOU)-**
Formula= Intersection area / Total area
If the IOU is greater than a threshold value (Eg- 0.7) then we conclude that the object is present in that bounding box.